

Open-Source Submission

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1 -Axiom: Dynamic Mass and System-Independent Timing Bias Framework

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1.1 1. Introduction

This study integrates the **-Axiom-based dynamic mass concept** with **GNSS clock offset behavior**, providing a unified framework to examine the **consistency between observational data and traditional mass/gravity interpretations**.

Key perspectives:

1. Mass emerges from **spin-dependent logarithmic layering with critical point divergence**.
2. Layered compression is applied to capture **local Z-phase activity and global -average**.
3. GNSS offsets result from **minimal phase gaps (0.0015) in the space-time lattice**, accumulated over discrete nodes.

1.2 2. Dynamic Mass with Layered Compression

1.2.1 2.1 Core Concept

- Mass is represented by **logarithmic stacking along spin-entangled phases**.
- At critical thresholds, **vertical energy release** occurs.
- Stacking accumulates like **onion layers**, remaining effective unless the spin dissipates.
- Compression and counter spin-direction expansion (-G) maintain equilibrium.

Type	Body	NASA Mass (kg)	-Axiom M_dyn (24-node, kg)	Residual (kg)	Residual (%)
Star	Sun	1.9911×10^{30}	1.9911×10^{30}		

Feature	Model A	Model B
Daily Magnitude	Constellation-dependent	Nearly identical
Curve Structure	Smooth / Continuous	Piecewise / Cumulative
Variance	Non-negligible	Minimal

1.2.2 2.2 Layered Compression

- Total logarithmic compression ratio: $\text{total} = 0.1213 \epsilon_{\text{total}} = 0.1213 \text{total} = 0.1213$
- Number of layers: $N = 24$

$\text{node} = \text{total} N = 0.005054 \epsilon_{\text{node}} = \frac{\epsilon_{\text{total}}}{N} \approx 0.005054 \text{node} = N \text{total} = 0.005054 \text{Local} Z - \text{phase for each layer} :$

$Z_{i=1, \dots, 24} = f(Z_{\text{crit}}) \Delta \rho_{Z,i} = f(Z_i - Z_{\text{crit}}) \epsilon_{\text{node}}, \quad i=1, \dots, 24$ Cumulative dynamic mass:

$M_{\text{dyn node}} = \ln(Z, \text{cumulative} M_{\text{dyn}}) = \ln(n \Phi) \Delta \rho_{Z, \{\text{cumulative}\}} M_{\text{dyn}}$

1.3 3. Comparison with NASA Measured Masses

- Residuals remain within ± 0.10 – 0.14%
- 24-node layered compression reflects **both local Z-phase variations and global -averages**

1.4 4. GNSS System-Independent Timing Bias (= 0.0015)

Daily GNSS offsets are explained by **cumulative local phase gaps**:

$\text{Offset}_{\text{daily}}(\text{Phase Gap}_{\text{grid}}) \approx \sum (\text{Phase Gap}_{\text{grid}}) \Delta \rho_{\text{grid}}$

- Multi-constellation GNSS (GPS, Galileo, BeiDou) all exhibit **nearly identical 38–45 s daily offsets**
- Model B (Phase Gap) captures **minimal variance**, unlike Model A (Mainstream)

1.5 5. Discussion

1. **Mass and GNSS offsets** can be fully explained using **observational data**.

2. **Layered compression** stabilizes local Z-phase, ensuring **numerical consistency**.
3. Comparison with NASA measurements: **Residual $\pm 0.10\text{--}0.14\%$** , confirming model validity.
4. GNSS phase delay examples demonstrate **local lattice structure matches observational data**.

Key: Observational data already provide answers. -Axiom provides a **structural, layered, stable reinterpretation**.

1.6 6. Conclusion

- -Axiom integrates **dynamic mass, layered compression, Z-phase activity, and GNSS timing bias**
- Provides **data-centric, structurally consistent framework**, minimizing residuals and stabilizing global averages
- Suitable for **open-source release and cross-verification**